



Securitization Dashboard

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Project: Project 1A – Securitisation Risk Assessment Dashboard

Tool Used: Microsoft Power BI

Submission Date: July 2026

1. Project Overview and Objectives

- a. The objective of this project is to develop an interactive Power BI dashboard for analysing auto loan securitisation portfolios. The dashboard enables users to monitor loan performance, credit quality, portfolio exposure, delinquency trends, and expected credit losses using DAX calculations and interactive visualisations.
- b. The solution helps financial institutions, investors, and risk managers evaluate portfolio health and make informed lending and investment decisions.

2. Objectives

The main objectives of this project are:

- Build an interactive Power BI dashboard.
- Analyses loan portfolio performance.
- Monitor IFRS 9 Stage classification.
- Analyses delinquency (DPD) buckets.
- Track Expected Credit Loss (ECL).
- Evaluate borrower credit quality using CIBIL scores.
- Provide interactive filtering using slicers.
- Create reusable DAX measures for portfolio analysis.

3. TOOLS AND TECHNOLOGIES

- Power BI
- DAX
- Excel
- CSV Datase

4. Dataset Description

The project uses an Auto Loan Securitisation dataset containing borrower, loan, and portfolio information.

Key fields include:

- Borrower ID

- Borrower Age
- Current Balance
- Interest Rate
- CIBIL Score
- DTI Ratio
- IFRS Stage
- DPD Bucket
- Employment Type
- Vehicle Type
- Insurance Type
- EAD
- ECL Provision
- Default Status

5. Dashboard Components

The dashboard includes the following visualizations:

KPI Cards

- Outstanding Portfolio Balance
- Total EAD
- Average Interest Rate
- Average CIBIL Score
- Average Borrower Age
- Average DTI Ratio
- Total Borrowers
- Total ECL Provision

Charts

- Borrower Age Distribution
- Portfolio Balance by IFRS Stage
- Portfolio Balance by DPD Bucket

Interactive Slicers

- IFRS Stage
- DPD Bucket
- Employment Type
- Vehicle Type

- Insurance Type
- Default Status

6. DAX Measures Implemented

The following DAX measures were created:

Total Portfolio Balance

Total Portfolio Balance =
SUM(auto_loan_securitisation_data[CurrentBalance])

Total EAD

Total EAD =
SUM(auto_loan_securitisation_data[EAD])

Average Interest Rate

Average Interest Rate =
AVERAGE(auto_loan_securitisation_data[InterestRate])

Average Borrower Age

Average Borrower Age =
AVERAGE(auto_loan_securitisation_data[BorrowerAge])

Average CIBIL Score

Average CIBIL Score =
AVERAGE(auto_loan_securitisation_data[CIBIL_Score_Current])

Average DTI Ratio

Average DTI Ratio =
AVERAGE(auto_loan_securitisation_data[DTI_Ratio])

Portfolio ECL

Portfolio ECL =
SUM(auto_loan_securitisation_data[ECL_Provision])

Total Borrowers

Total Borrowers =
COUNT(auto_loan_securitisation_data[BorrowerID])

5. Dashboard Development

Dashboard Overview

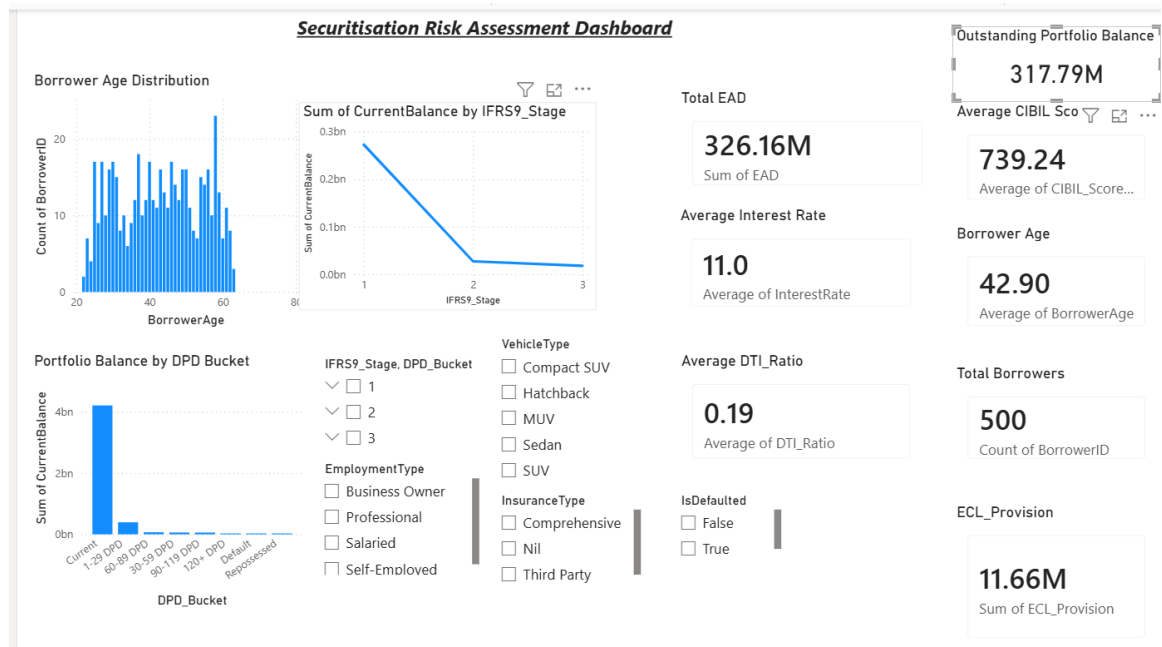
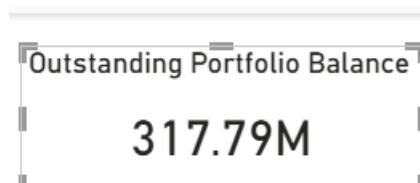


Figure 1. Securitization Risk Assessment Dashboard

The dashboard provides an overview of the loan portfolio, borrower profile, credit quality, expected credit losses, and delinquency status. Interactive filters allow users to analyse risk across multiple borrower and loan characteristics.

KPI CARDS



Outstanding Portfolio Balance – Displays the total active loan balance

Total EAD

326.16M

Sum of EAD

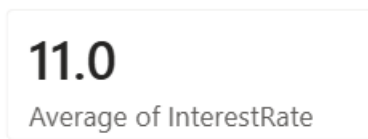
Total EAD – Shows the Exposure at Default.

Average CIBIL Score



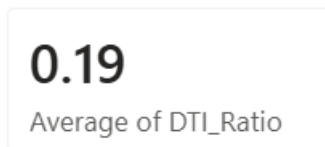
Average CIBIL Score – Represents portfolio credit quality.

Average Interest Rate



Average Interest Rate – Average lending rate.

Average DTI_Ratio

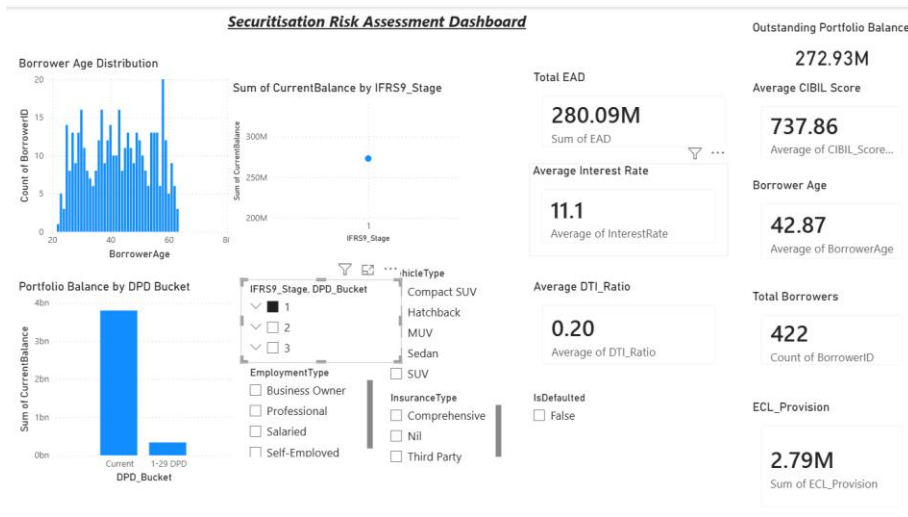


Average DTI Ratio – Indicates borrower leverage.

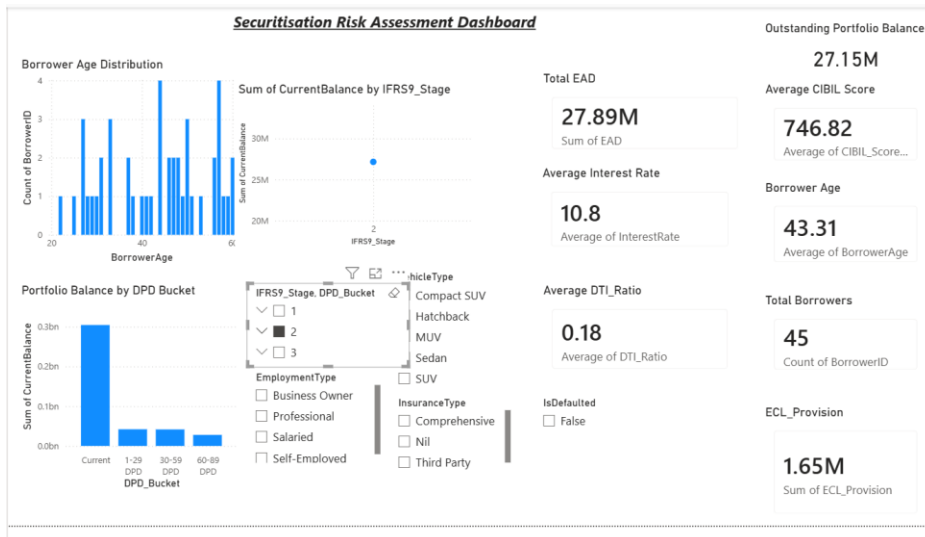


ECL Provision – Estimated Expected Credit Loss.

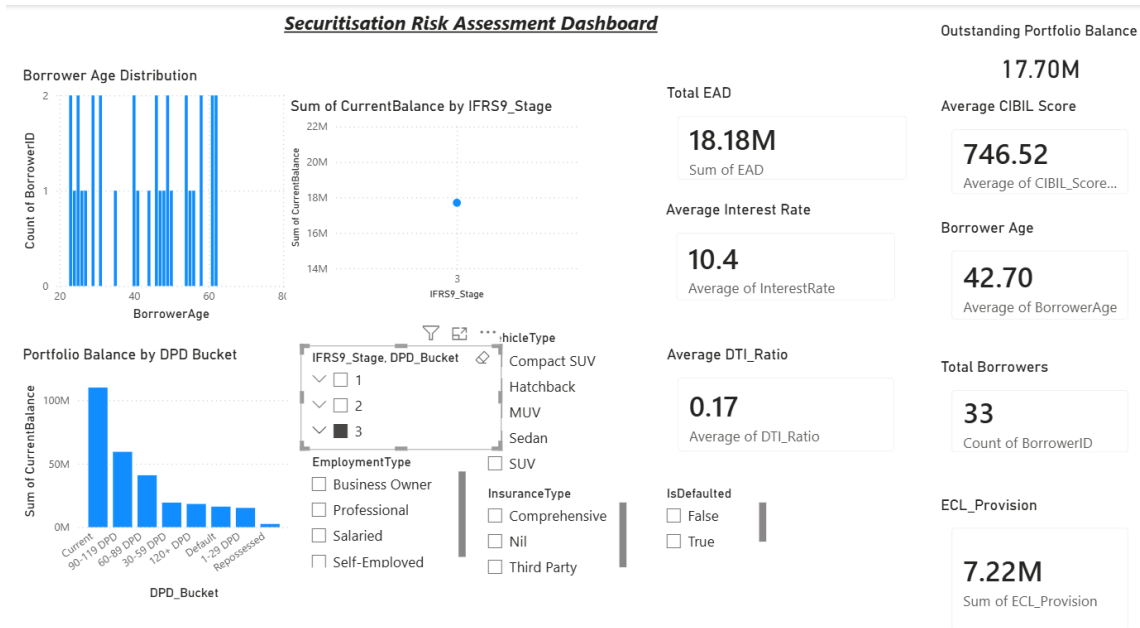
IFRS Stage Analysis:



Stage 1



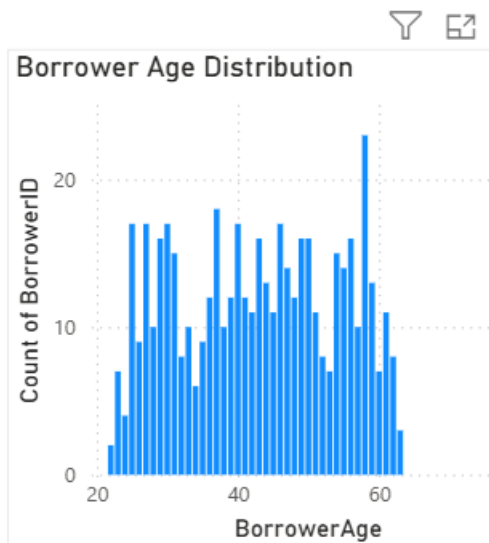
Stage 2



Stage 3

Stage	Status	Credit Risk Profile	ECL Recognition
Stage 1	Performing	No significant increase in credit risk since initial recognition.	12-month ECL: The portion of lifetime losses resulting from default events possible within the next 12 months.
Stage 2	Underperforming	Significant increase in credit risk since initial recognition, but not yet credit-impaired.	Lifetime ECL: The expected losses resulting from all possible default events over the expected life of the instrument.
Stage 3	Non-performing	Credit-impaired (objective evidence of impairment).	Lifetime ECL: Same as Stage 2, but interest revenue is calculated on the net carrying amount (net of the allowance).

Borrower Age Distribution



Shows borrower concentration across age groups, helping understand demographic risk

SLICERS:

VehicleType
☐ Compact SUV
☐ Hatchback
☐ MUV
☐ Sedan
☐ SUV

EmploymentType
☐ Business Owner
☐ Professional
☐ Salaried
☐ Self-Employed

InsuranceType
☐ Comprehensive
☐ Nil
☐ Third Party

IsDefaulted
☐ False
☐ True

Slicers enable dynamic portfolio analysis.

6. Key Findings

The dashboard provides valuable insights into the loan portfolio.

- Portfolio exposure is concentrated in IFRS Stage 1 loans.
 - Current loans represent the largest share of outstanding balances.
 - Borrowers have generally good credit quality based on average CIBIL scores.
 - DPD bucket analysis identifies delinquent loans requiring attention.
 - ECL Provision helps estimate expected credit losses.
 - Interactive slicers enable detailed portfolio analysis by borrower and loan characteristics.
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7. Business Benefits

The dashboard helps financial institutions to:

- Monitor portfolio health.
 - Identify high-risk borrowers.
 - Support IFRS 9 reporting.
 - Improve credit risk management.
 - Enhance investor reporting.
 - Enable data-driven lending decisions.
 - Simplify portfolio monitoring through interactive visualizations.
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8. Challenges Faced

During development, several challenges were addressed:

- Understanding relationships between loan attributes.
 - Creating accurate DAX measures.
 - Selecting appropriate visualizations.
 - Designing an easy-to-understand dashboard layout.
 - Validating calculated measures against source data.
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9. Conclusion

The Securitisation Risk Assessment Dashboard successfully transforms raw loan data into meaningful business insights. Using Power BI and DAX, the dashboard provides an interactive platform for monitoring portfolio performance, analysing credit risk, and supporting informed decision-making.

The project demonstrates practical skills in data modelling, DAX calculations, dashboard development, and financial risk analysis.

10. Future Enhancements

Future improvements may include:

- Integration with live databases such as Azure SQL or SAP.
- Real-time dashboard refresh.
- Predictive analytics using Machine Learning.
- Automated risk alerts.
- Advanced stress testing scenarios.
- Portfolio forecasting using AI models.